

Assignment 03 [SKZ | Summer2026]

Total points 60/100 ?

Transport Layer - UDP and TCP

The respondent's email (**md.rimon.hossain2@g.bracu.ac.bd**) was recorded on submission of this form.

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✓ Which of the following is a **primary function of the transport layer?** 5/5

- ☐ Routing packets between routers
- ☒ Logical communication between processes
- ☐ Assigning IP addresses
- ☐ Managing physical transmission

✓

✓ The TCP and UDP common field/s of header is/are 5/5

- ☐ Header Length
- ☒ Port Addresses
- ☐ Sequence Number
- ☒ Checksum

✓

✓



✓ Which of the following contributes to **UDP's low delay**?

5/5

- ☐ Large header size
- ☐ Connection establishment
- ☒ No congestion and flow control
- ☐ Multiple retransmissions



✓ In **UDP demultiplexing**, which field is primarily used?

5/5

- ☐ Source IP
- ☒ Destination port number
- ☐ Sequence number
- ☐ Window size



✓ **Checksum** in UDP is mandatory

Dropdown 5/5

1. TRUE

2. FALSE



A server just received an acknowledgement segment from the client containing sequence number of 501 and acknowledgement number of 4000. Now the server will send two segments of the size 500 bytes to the client. The client replies with an acknowledgement segment.

Acknowledgement number: 5000



✗ Refer to the above question, what is the sequence number of the ack segment sent by the client? Just write the number.

0/5

501

✗

Correct answer

5000

✗ Refer to the above question, what is the acknowledgement number of the last ack segment sent by the client? Just write the number. .../5

5000

✗

Correct answer

502

TCP Math

In a TCP connection, the initial sequence number at the client site is 6710 with rwnd of 7000 and the initial sequence number at the web server is 355 . The client initiates the three way handshake, and with the third segment of the three way handshake requests (HTTP Request of size 3636) for 6 objects, each of which carries 464 bytes of data. The server sends the requested objects using persistent HTTP connection with a Go-Back-N setup in the TCP layer. However, the 3rd packet gets lost on its way. What would be the value of the

✗ Question (a) acknowledgment number of the server's SYN segment

.../5

6711

✗

✗ Question (b) sequence number of the second data segment sent by the server .../5

820

✗

✗ Question 1 - (c) What would be the window size of the client when the client receives the 6th packet? .../10

6072

✗

✗ Question 1 - (d) acknowledgement number of the 6th data segment that the server sent. .../10

10347

✗

✓ The flag(s) responsible for marking a segment/data as important is/are: 5/5

☒ URG

✓

☐ ACK

☒ PSH

✓

☐ RST

☐ SYN

☐ FIN

✓ The flag(s) responsible for establishing a connection is/are :

5/5

☐ URG

☒ ACK

✓

☐ PSH

☐ RST

☒ SYN

✓

☐ FIN

✓ The flag(s) responsible for terminating a connection is/are :

5/5

☐ URG

☒ ACK

✓

☐ PSH

☐ RST

☐ SYN

☒ FIN

✓

✓ 400 bytes of window size of Client A means

5/5

- ☐ Client A can only receive 400 bytes one time
- ☐ Client A can only send 400 bytes
- ☒ Client A can receive 400 bytes unless otherwise specified by Client A
- ☐ Client A can receive 400 bytes in four segments

✓

✓ Two browser tabs from the same computer client connect to the same web server. How are the sessions distinguished by the client?

*5/5

- ☐ Different HTTP methods
- ☐ Different destination IP
- ☒ Different source port numbers
- ☐ Different protocols

✓

✓ Which of the following protocols sometimes runs over **UDP**?

4/4

- ☒ DNS
- ☐ FTP
- ☐ Telnet
- ☐ SMTP

✓

✓ TCP uses the following information to calculate the checksum

6/6

☒ TCP Header

✓

☐ Data Link header

☒ Pseudo IP header

✓

☒ TCP Data

✓

☐ IP Data

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